

Mustain Billah Taj

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PROFILE

Software Engineering student focused on backend development and building agentic, LLM-powered applications. I work mainly in Python and FastAPI, designing REST APIs and AI features backed by PostgreSQL, LangChain, and the Groq API, with an emphasis on clean service boundaries and reliable, production-minded systems.

EDUCATION

B.Sc. in Software Engineering

Islamic University of Technology (IUT)

Higher Secondary Certificate (HSC) & Secondary School Certificate (SSC)

Barishal Cadet College

TECHNICAL SKILLS

Languages	Python, Java, C++
Backend	FastAPI, REST APIs, WebSocket, WebRTC
AI / LLM	LangChain, Groq API, RAG, PGVector
Databases	PostgreSQL, MySQL, MongoDB
Tools	Docker, Git, Redis
Core CS	Data Structures, Algorithms, OOP

PROJECTS

Stream — Self-Hosted Pay-Per-View Live Streaming Platform

github.com/Mustain98/stream

Creators broadcast live over WebRTC and get paid for it, with free previews and Stripe checkout for full access.

- Built the WebRTC media server (SFU) from scratch with `aiortc` rather than using Mediasoup or LiveKit: signaling, track forwarding, rooms, and server-side preview cutoffs.
- Split business logic and media into two services linked only by a short-lived ticket, so the media server never handles credentials or payments.
- Integrated Stripe end to end: idempotent payments, automated creator payouts, and automatic refunds when a stream ends early.

J_Buddy — AI Job Copilot

github.com/Mustain98/J_buddy

A job-hunting assistant that reads your CV, finds real listings, ranks them by fit, and answers questions about them in chat.

- Parses the user's CV, generates its own job searches from it, and fetches live listings via a background ARQ worker to keep the API responsive.
- Scores each job against the CV by combining keyword overlap with embedding similarity, so matches surface even when the wording differs.
- Built a RAG chatbot grounded in the user's CV and job data, plus gap analysis, learning roadmaps, cover letters, and application tracking.

HealthHive — AI Personal Health Management Platform

github.com/Mustain98/Health-Hive | Live Demo

An AI coach drafts your plan and meals, but a human always approves it — and risky goals go to a real consultant.

- Conversational coach that builds a personalized plan — habits, calorie targets, meal structure — from goals, body metrics, and medical conditions.
- Generates meal plans by letting the LLM set per-meal constraints, retrieving candidates with PGVector, then assembling combinations deterministically to hit the macros.
- Designed so the AI drafts but never decides: nothing activates without user review, and risky goals are referred to a verified consultant, bookable over Agora video.

INTERESTS & EXTRACURRICULAR ACTIVITIES

- Member of the IUT Computer Society.
- Outside of code: football, volleyball, and travelling.